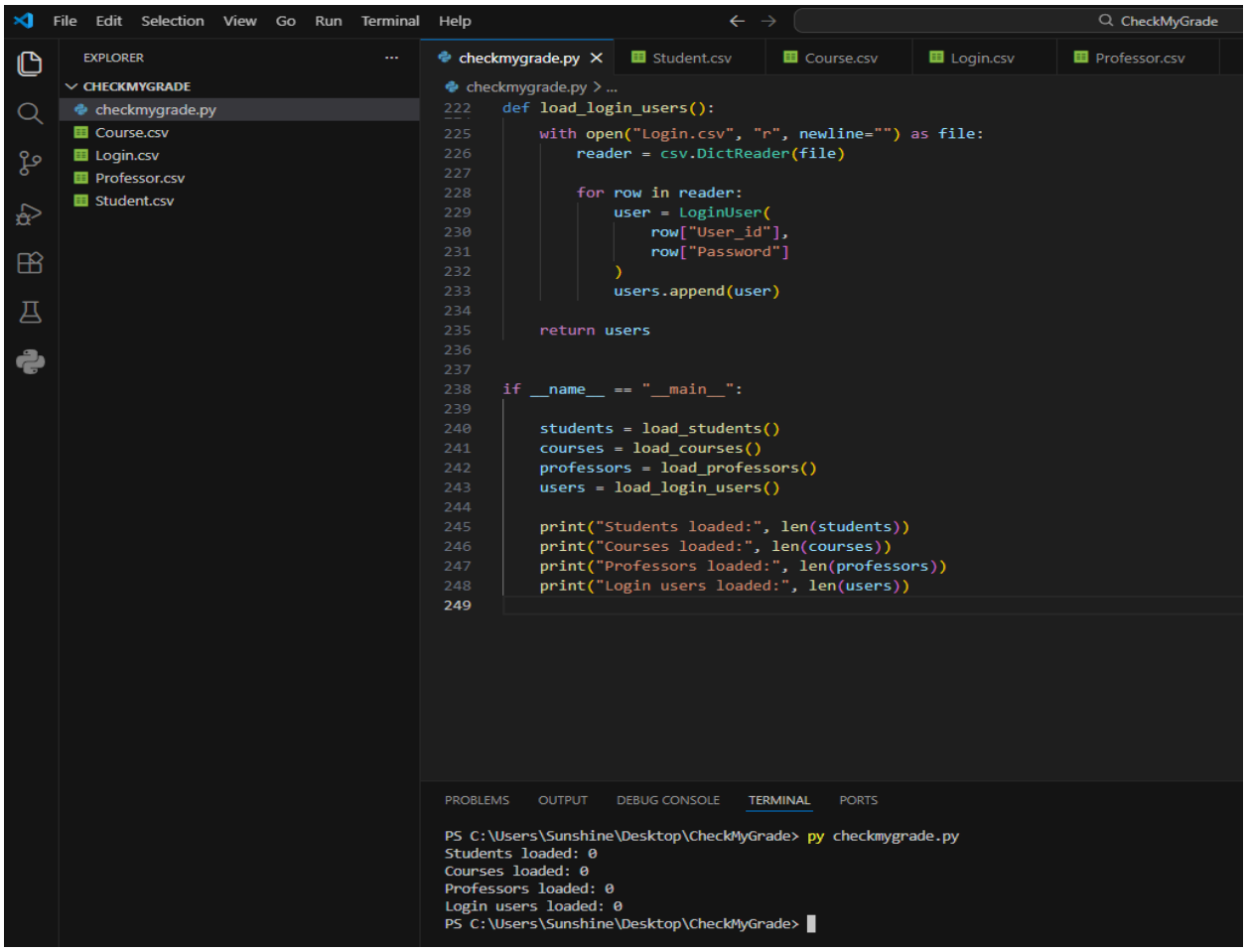
Screenshot Sections

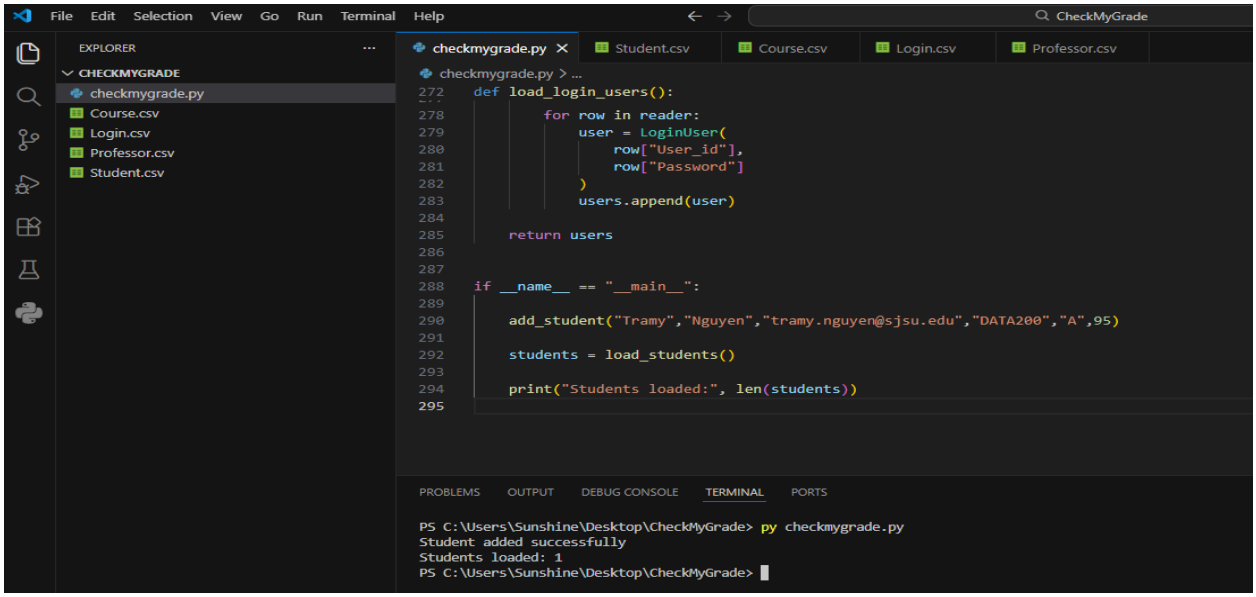
1. Program Setup/Running:

Running CheckMyGrade Application: This screenshot shows the CheckMyGrade Python application running successfully in the terminal.

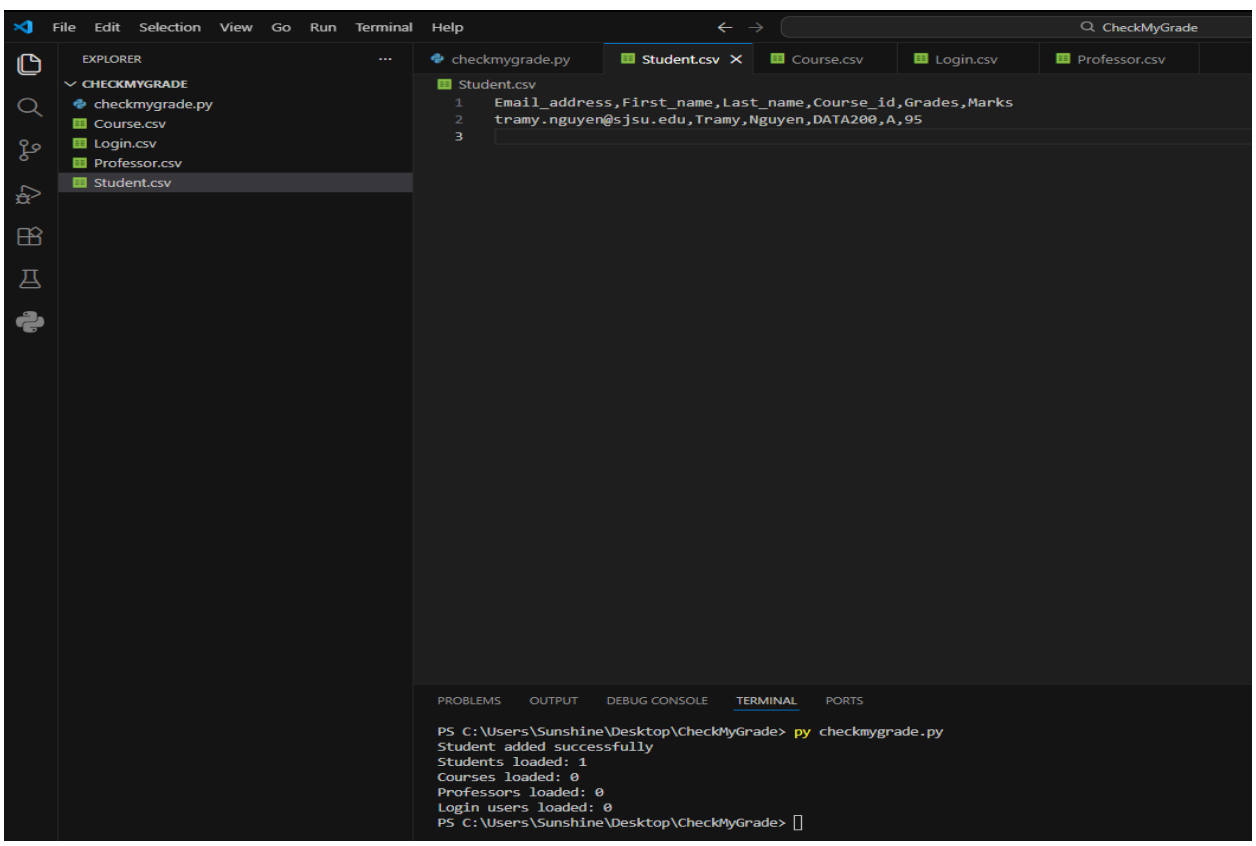


Screenshot Sections

2. Student Functions



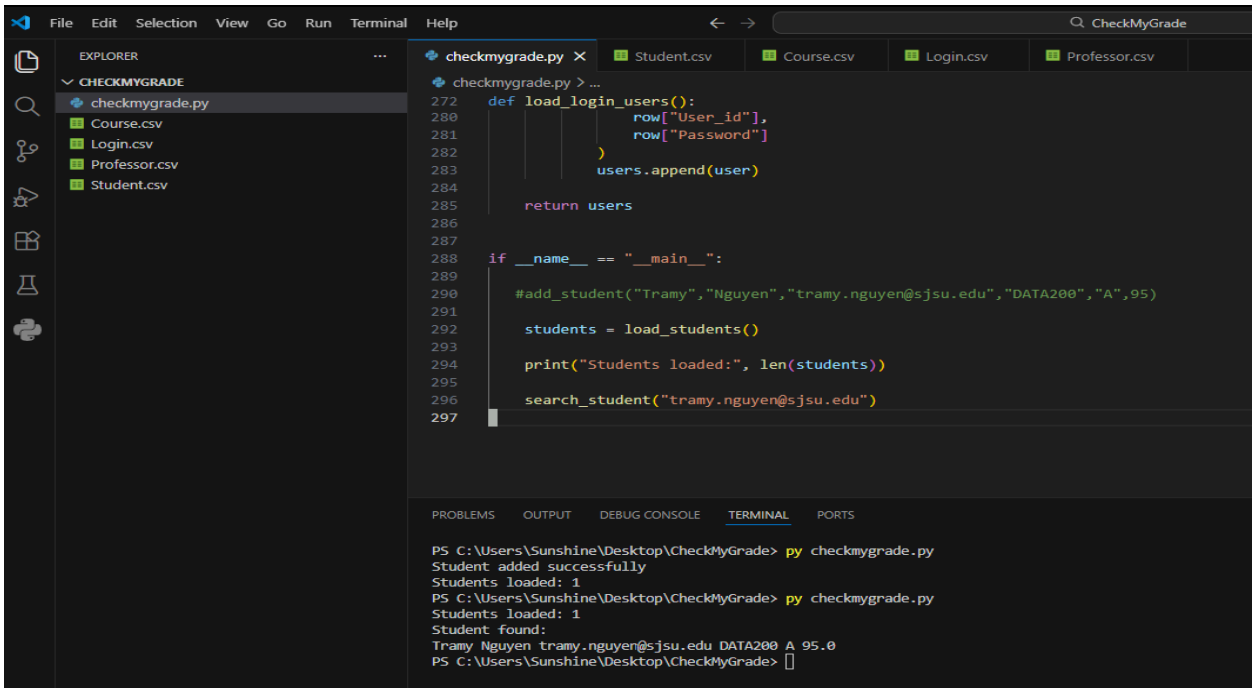
Adding a Student Record and Student Record Saved in CSV File: This function adds a new student record to the Student.csv file.



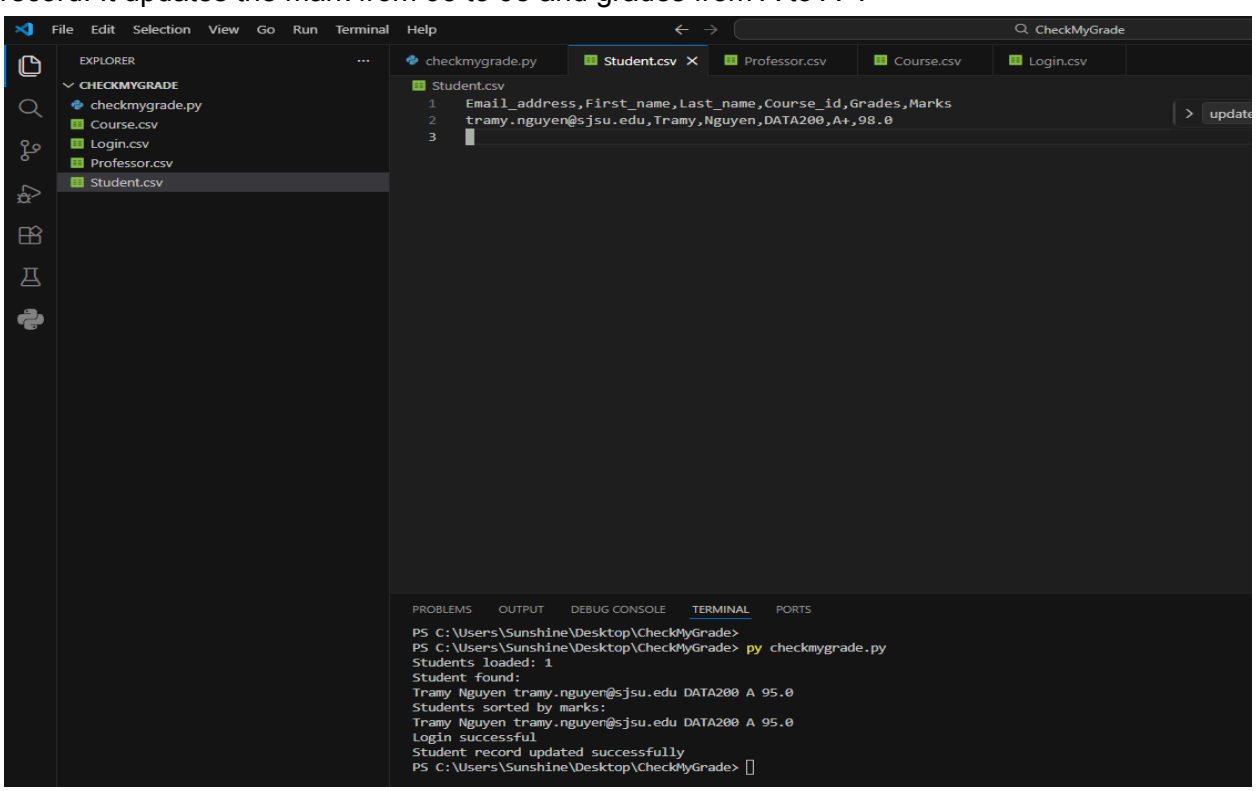
This screenshot confirms the student record was successfully stored in the Student.csv file.

Screenshot Sections

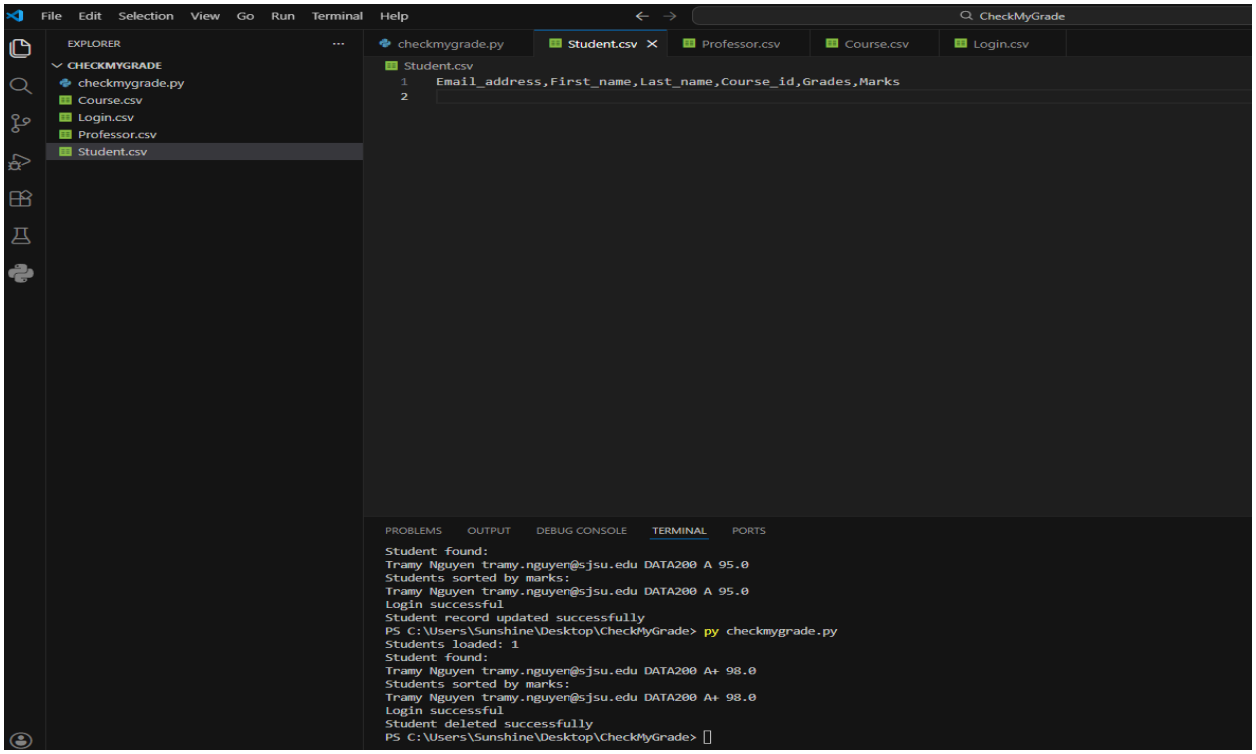
Searching for a Student: The search_student function retrieves a student record based on the email address.



Updating Student Record: This function updates the grade and marks of an existing student record. It updates the mark from 95 to 98 and grades from A to A+.

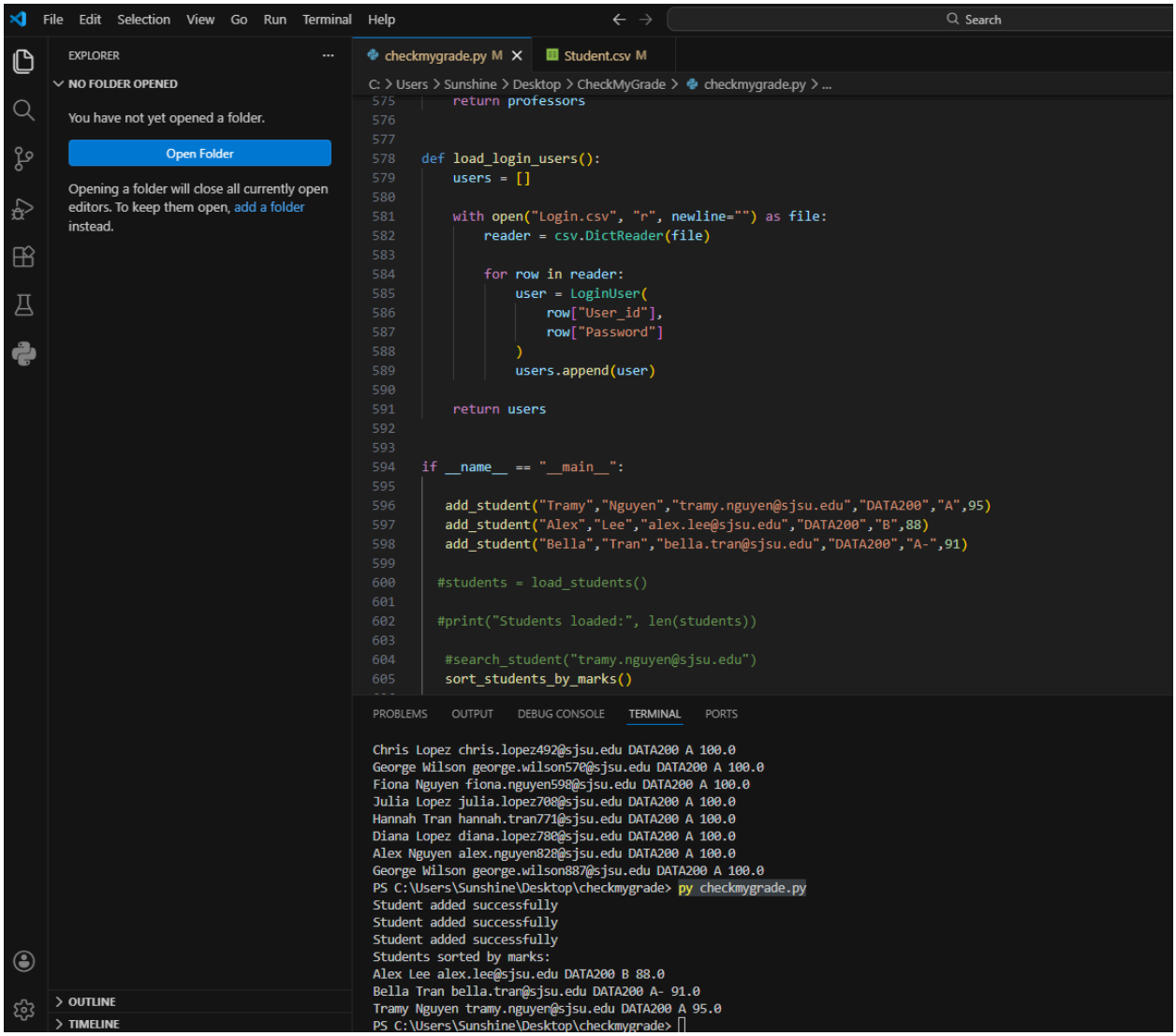


Screenshot Sections



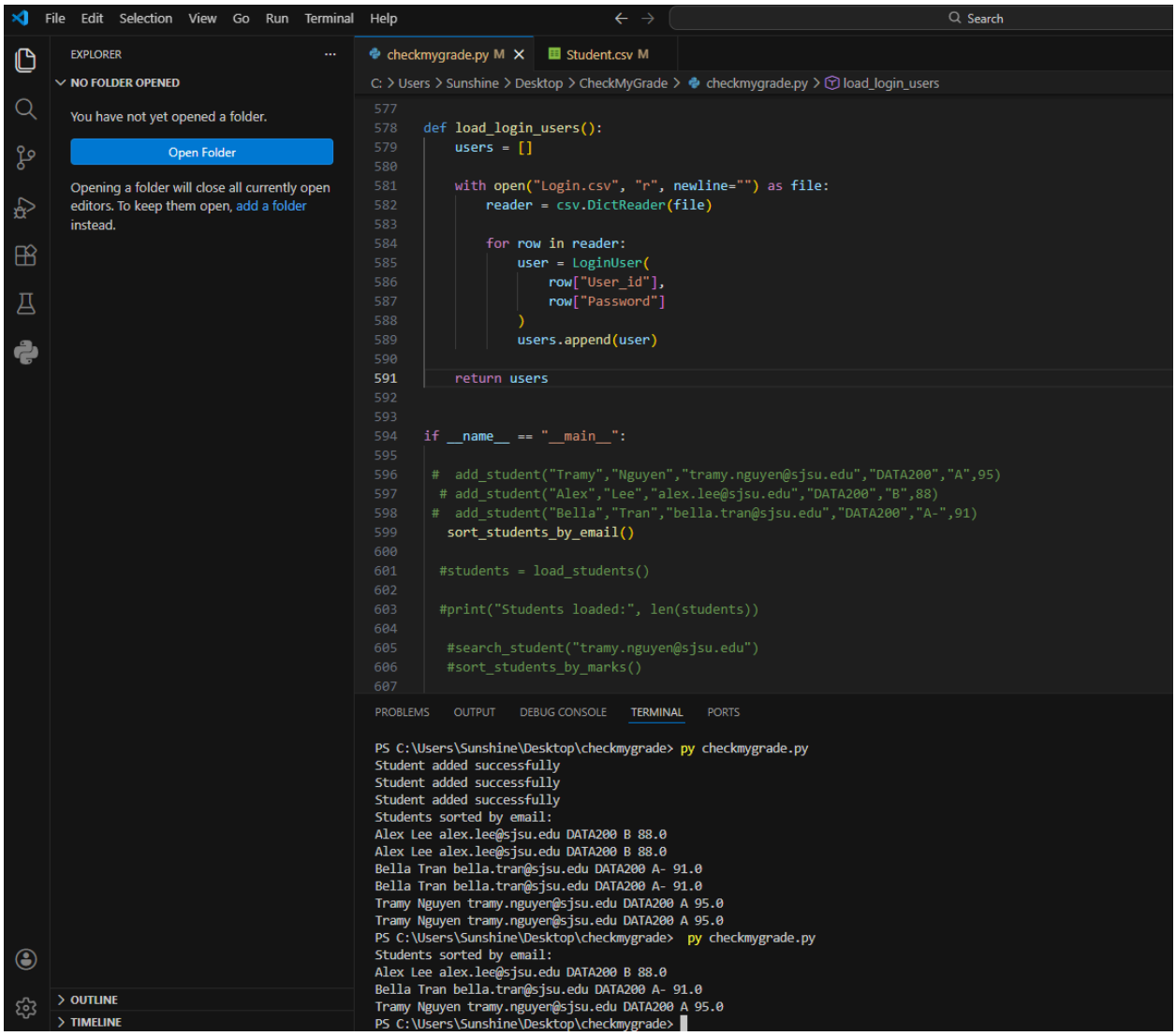
Deleting Student Record: This function removes a student record from the Student.csv file. It removes student Tramy Nguyen from the Student.csv.

Screenshot Sections



Sorting Students by Marks: Students are sorted in ascending order based on their marks.

Screenshot Sections

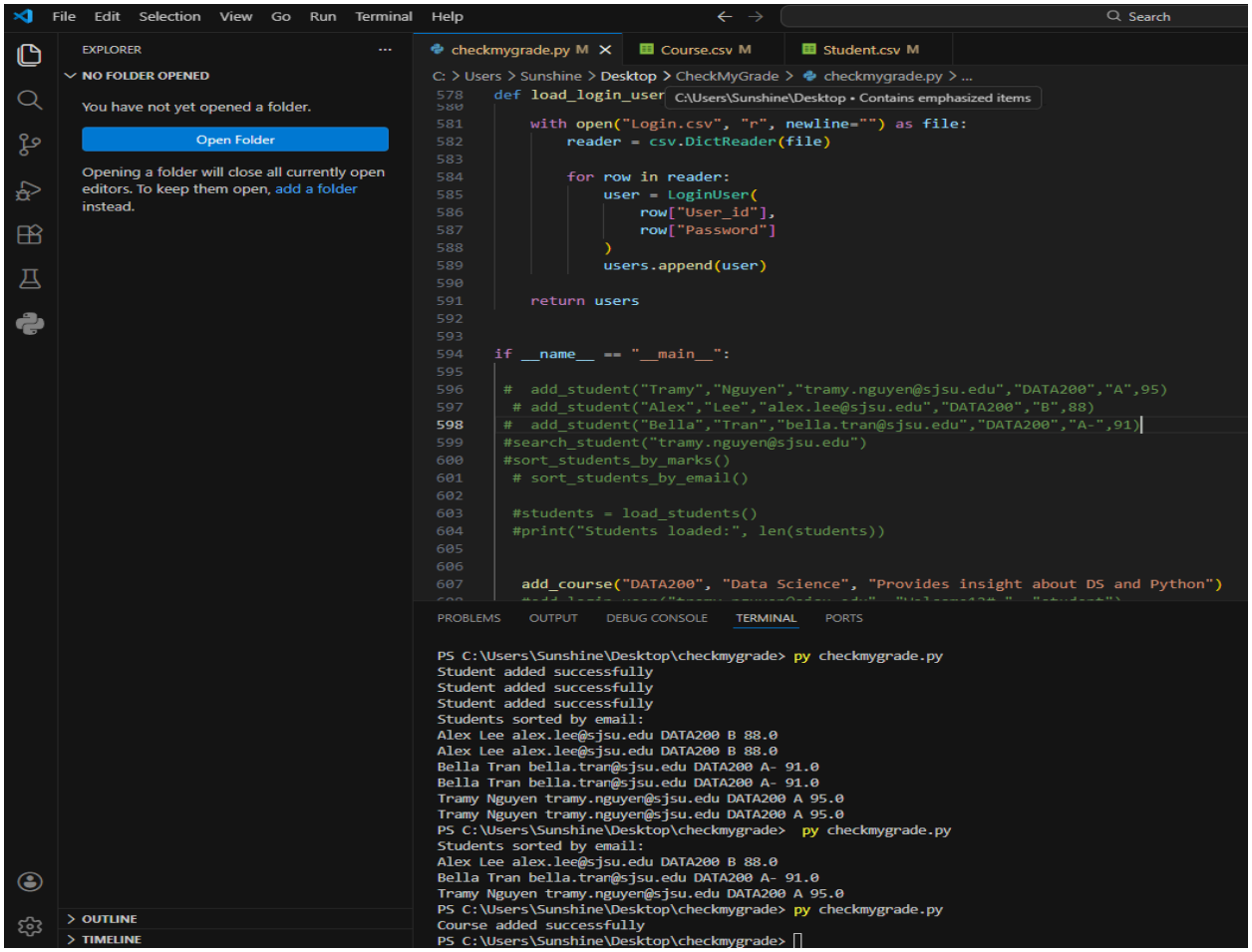


Sorting Students by Email Address: This function sorts students alphabetically based on their email addresses.

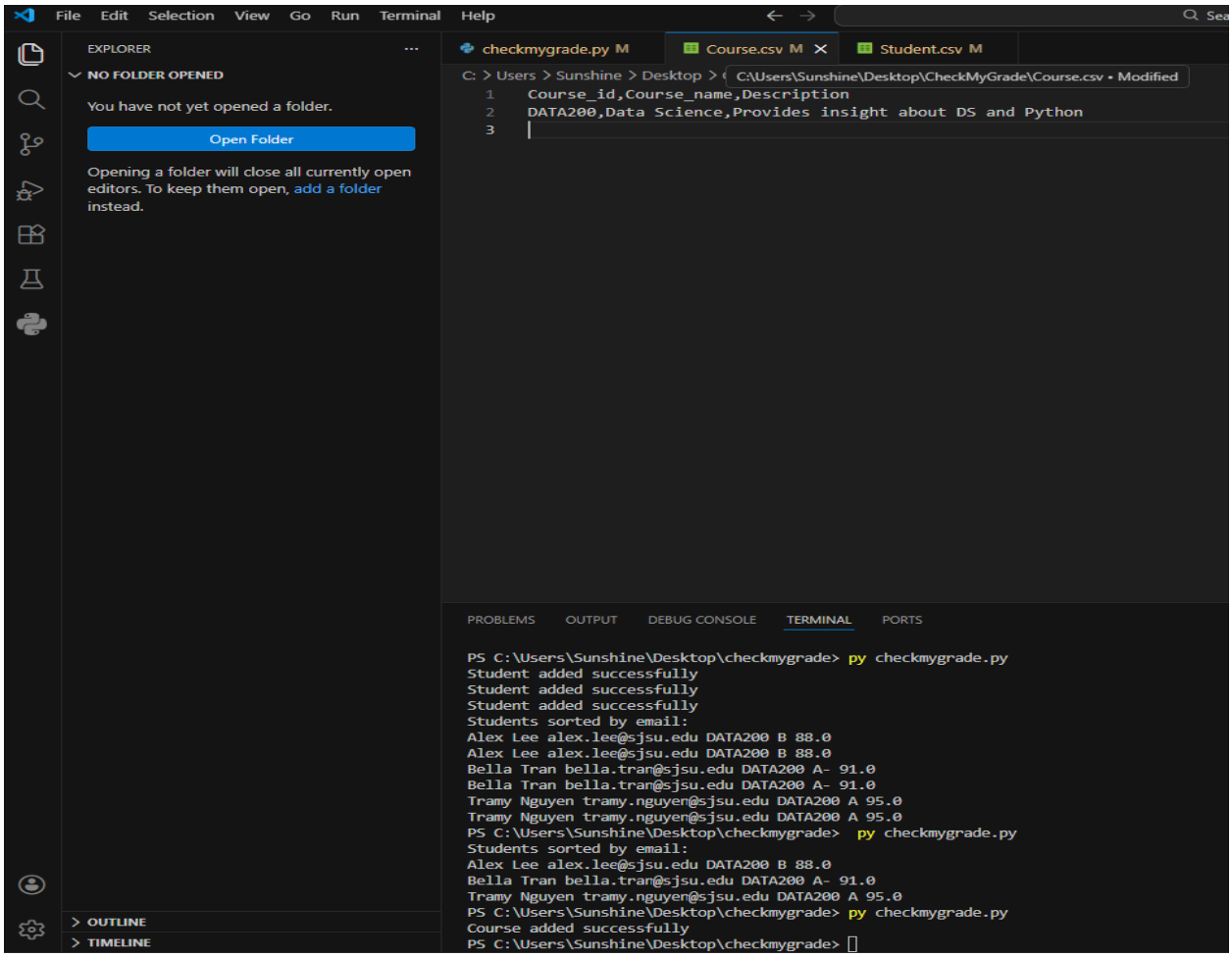
Screenshot Sections

3. Course Function:

Adding a Course: This function adds a new course record to the Course.csv file.

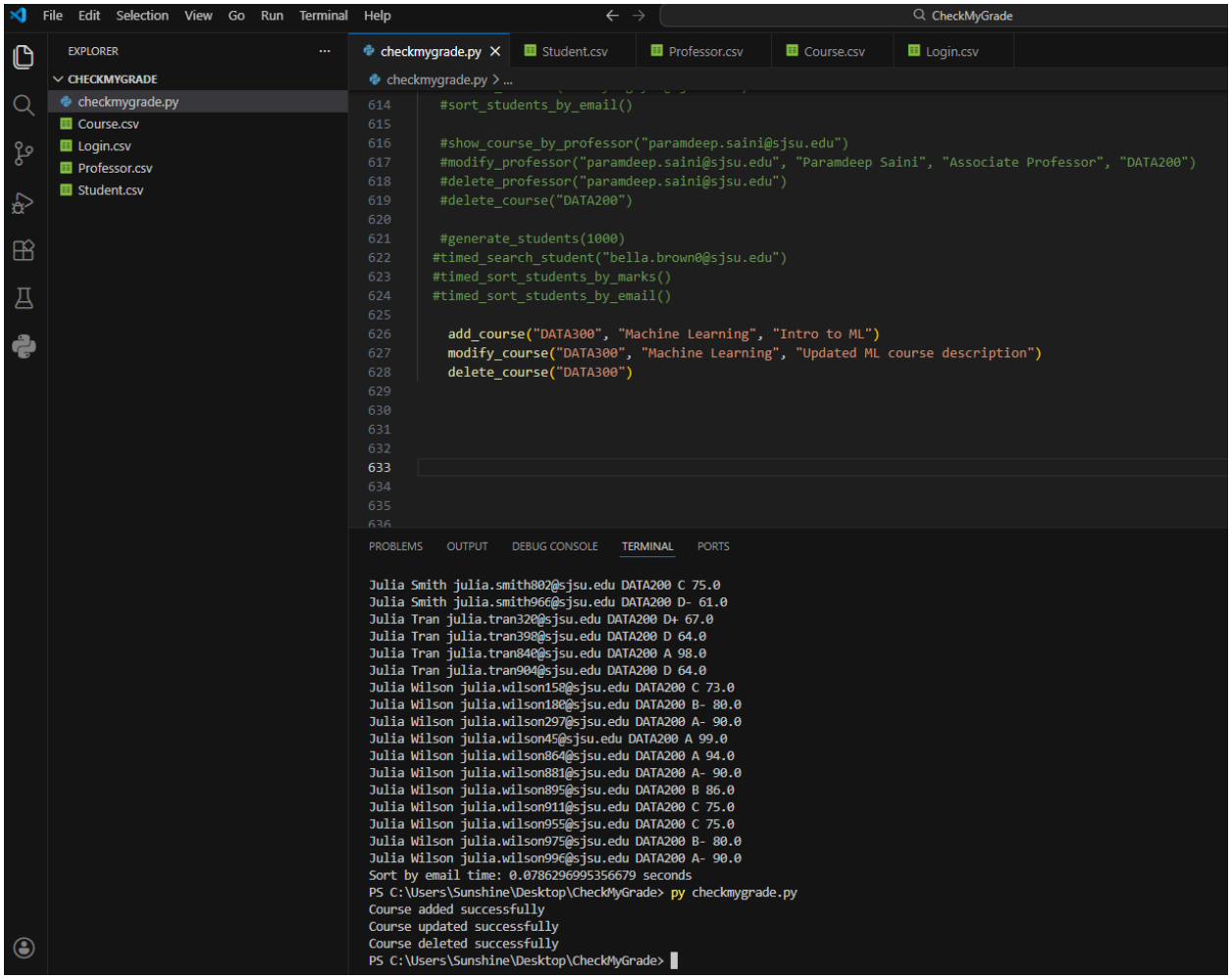


Screenshot Sections



This screenshot shows a new course is added to the Course.csv file.

Screenshot Sections

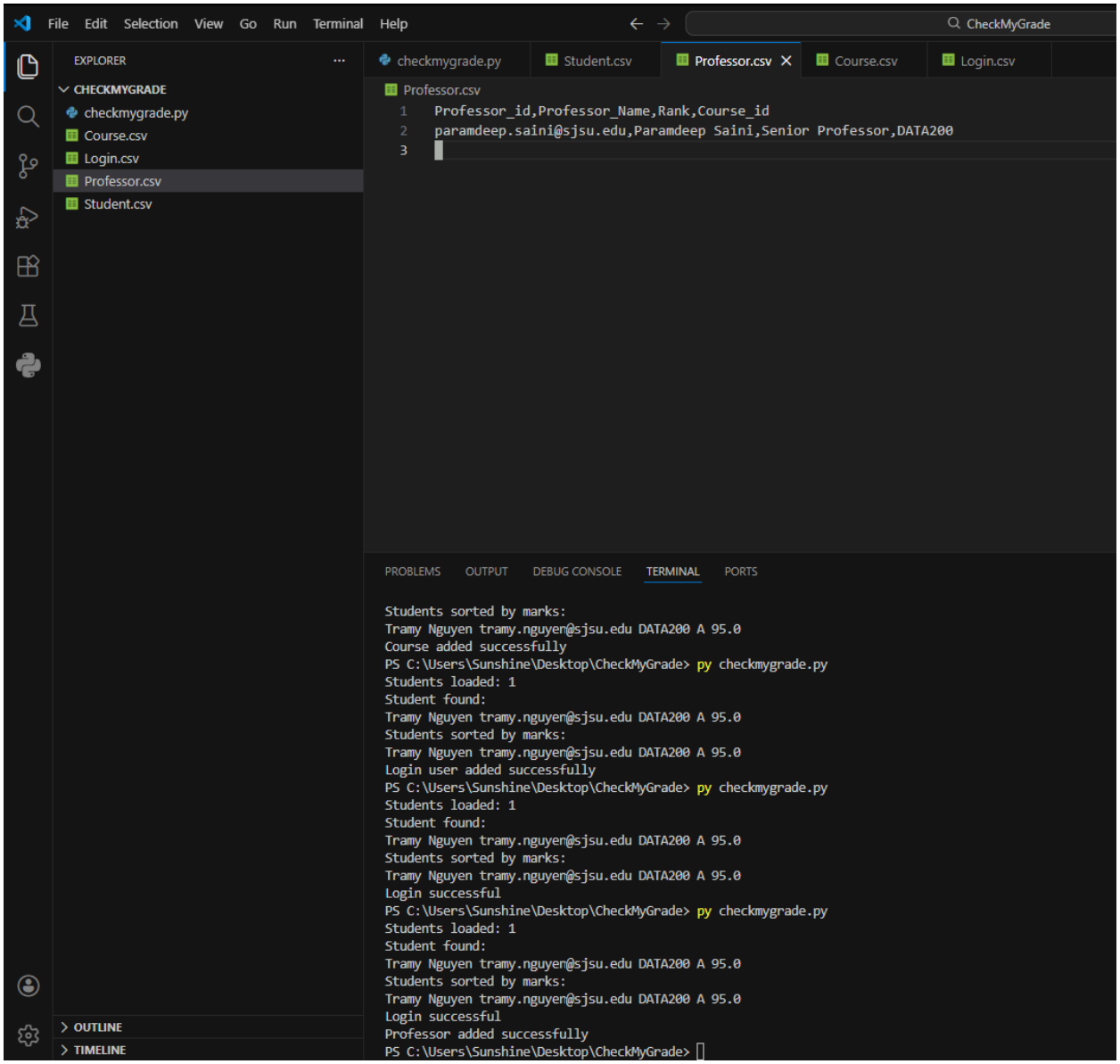


This screenshot shows a new course is added, and updated and then deleted.

Screenshot Sections

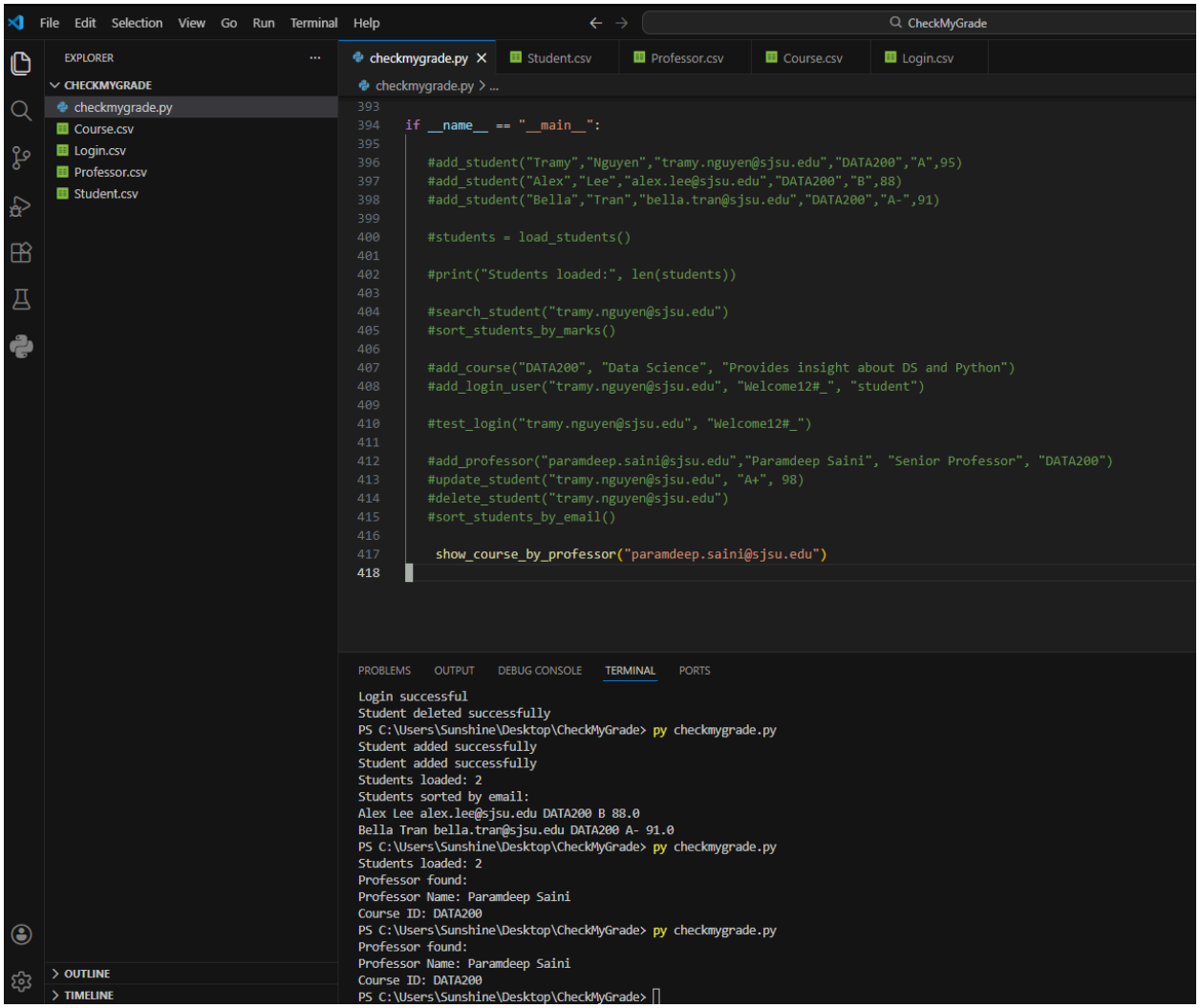
4. Professor Functions:

Adding a Professor: A new professor record is added to Professor.csv.



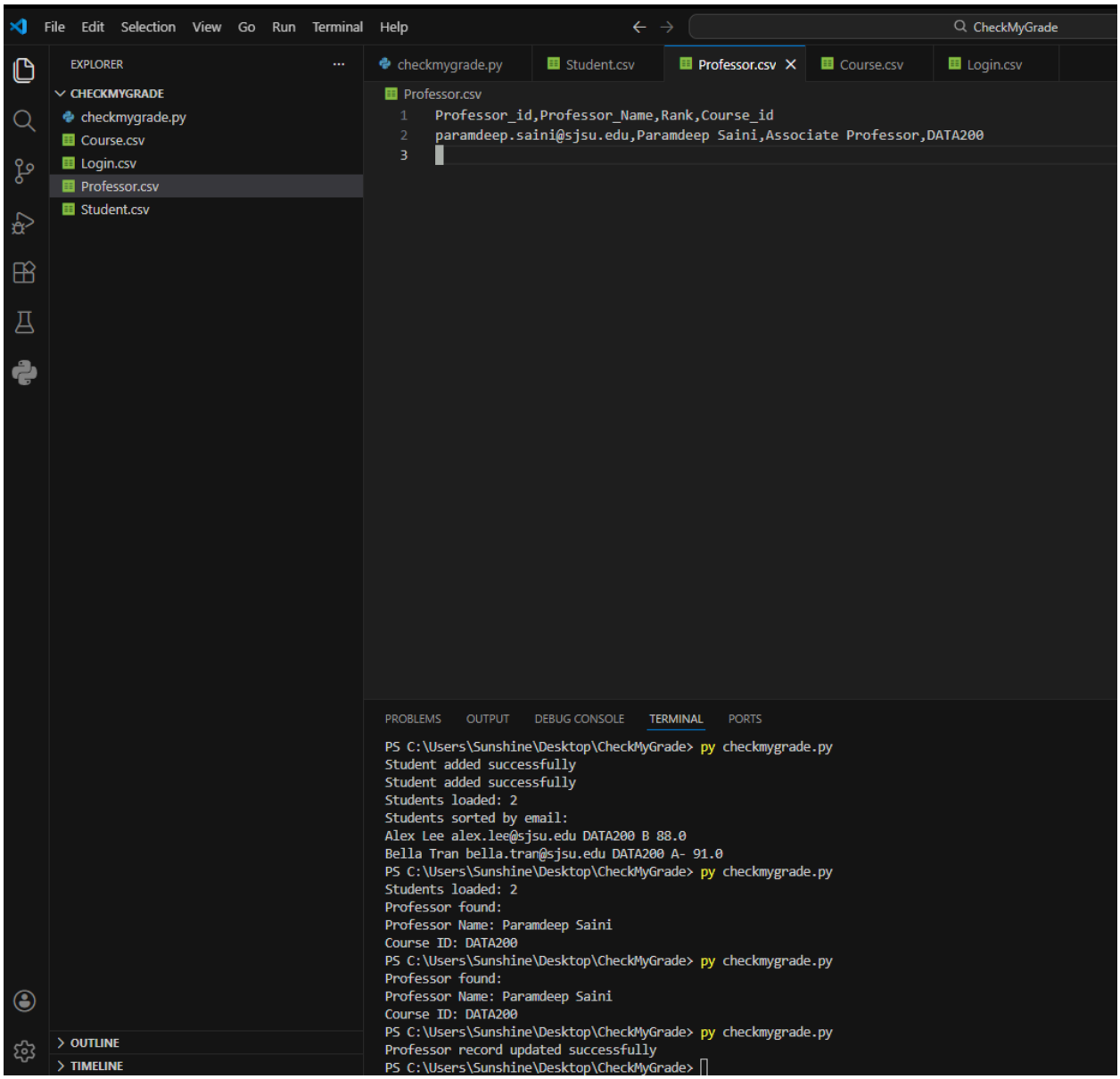
Screenshot Sections

Displaying Course by Professor: This function retrieves the course associated with a professor.



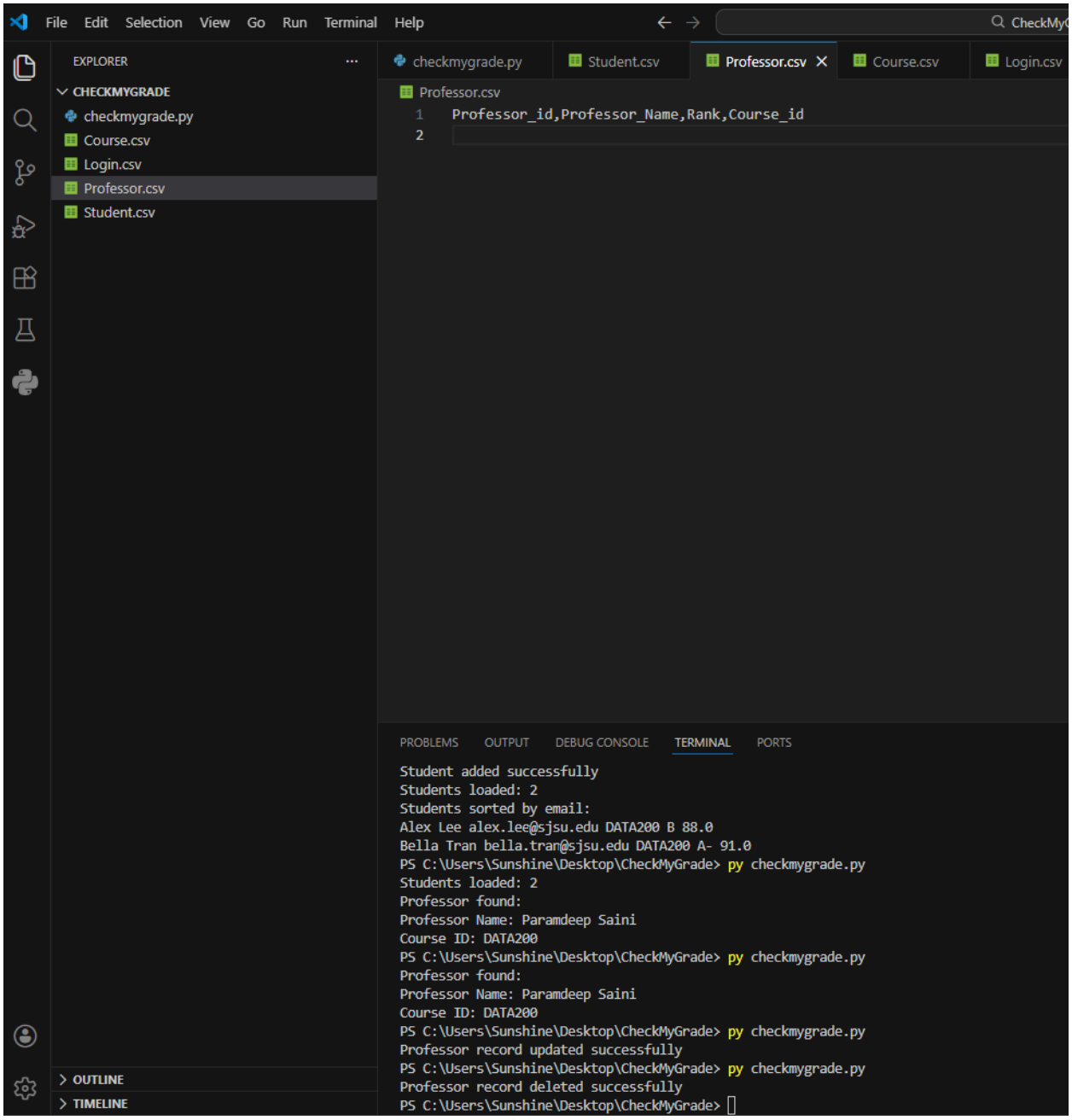
Screenshot Sections

Updating (Modify) Professor Record: The professor's rank or course assignment is updated (Senior Professor → Associate Professor)



Screenshot Sections

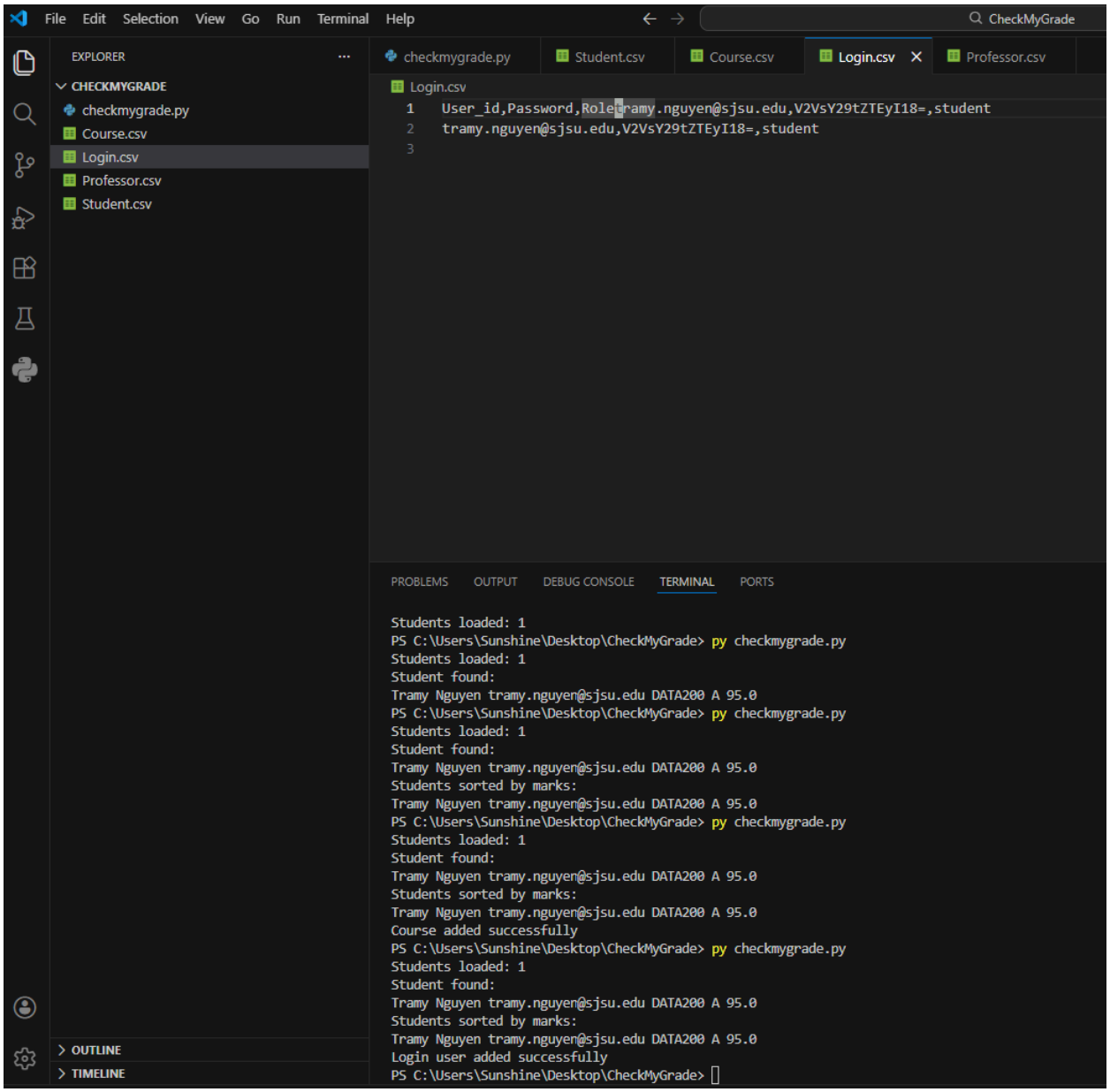
Deleting Professor Record: This function deletes the professor record from Professor.csv.



Screenshot Sections

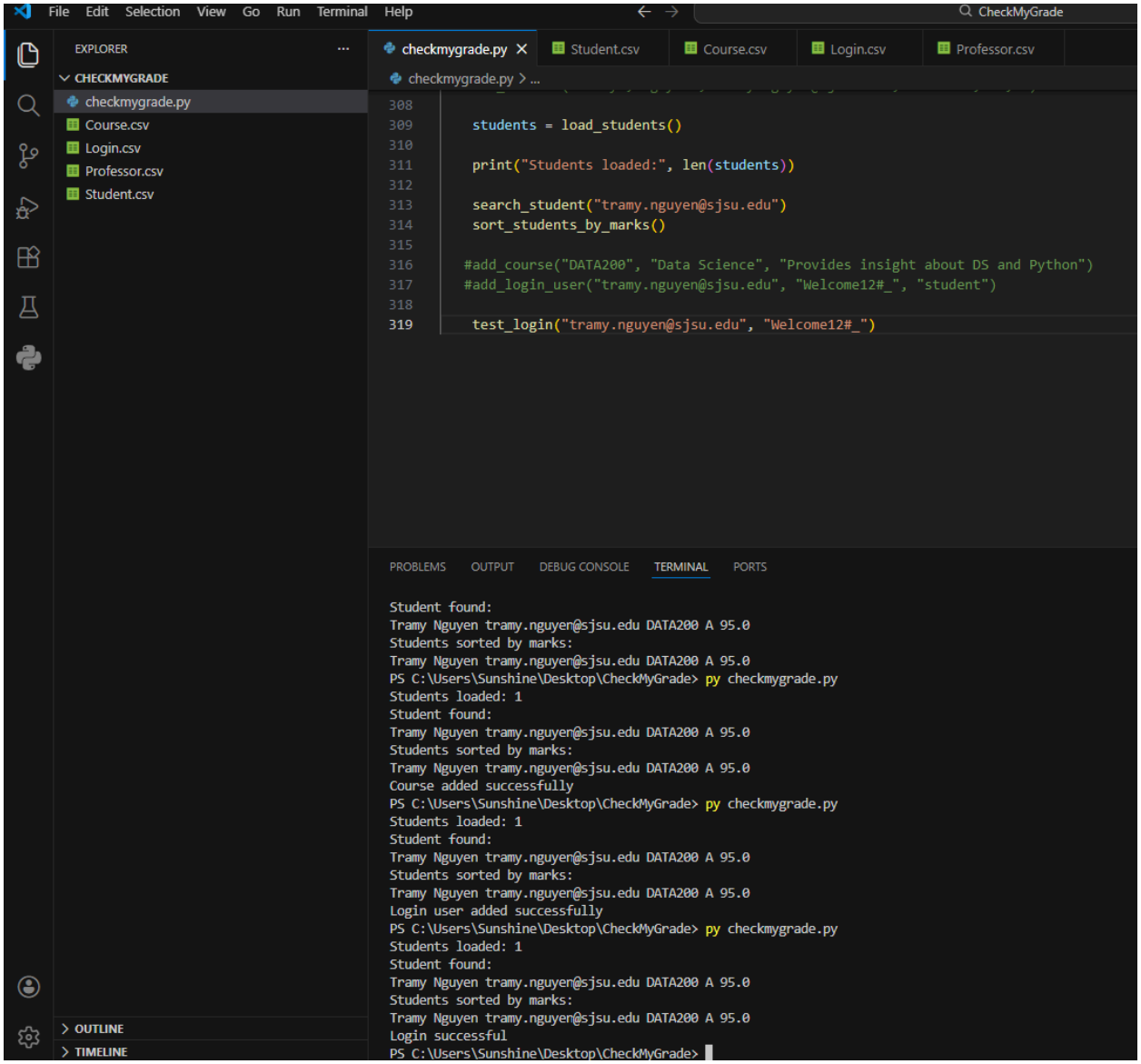
5. Login & Encryption

Adding Login User with Encrypted Password: The password is encrypted before being stored in the Login.csv file.



Screenshot Sections

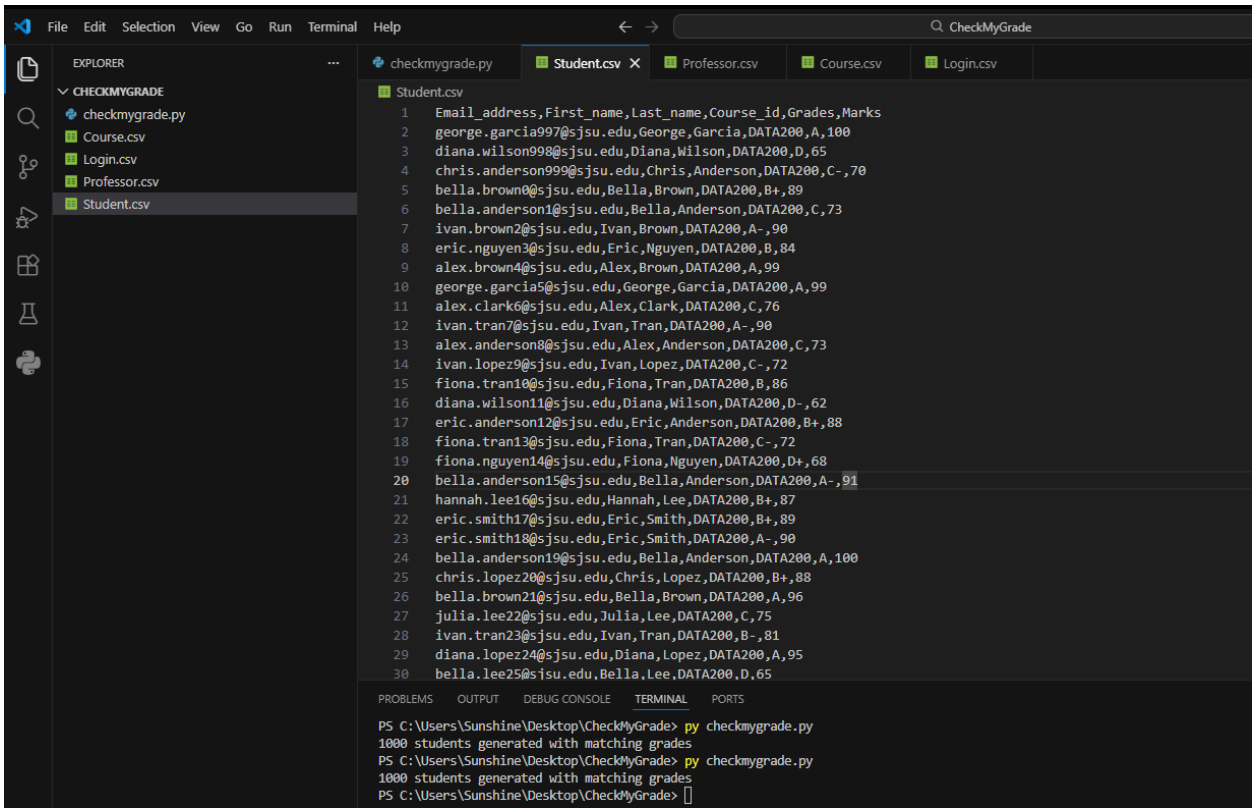
Testing Login Authentication: The system decrypts the stored password and validates the login credentials.



Screenshot Sections

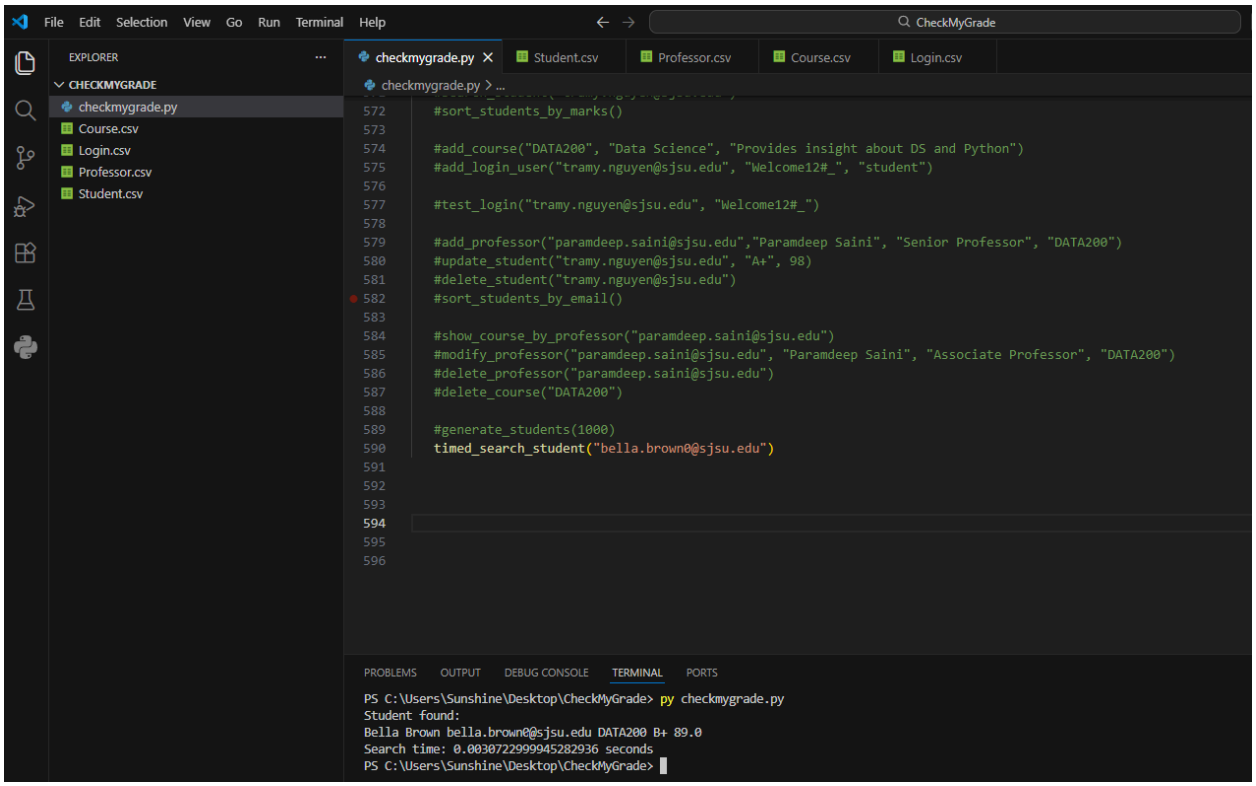
6. Unit Testing:

Generating 1000 Student Records: This function automatically generates 1000 student records for performance testing.



Screenshot Sections

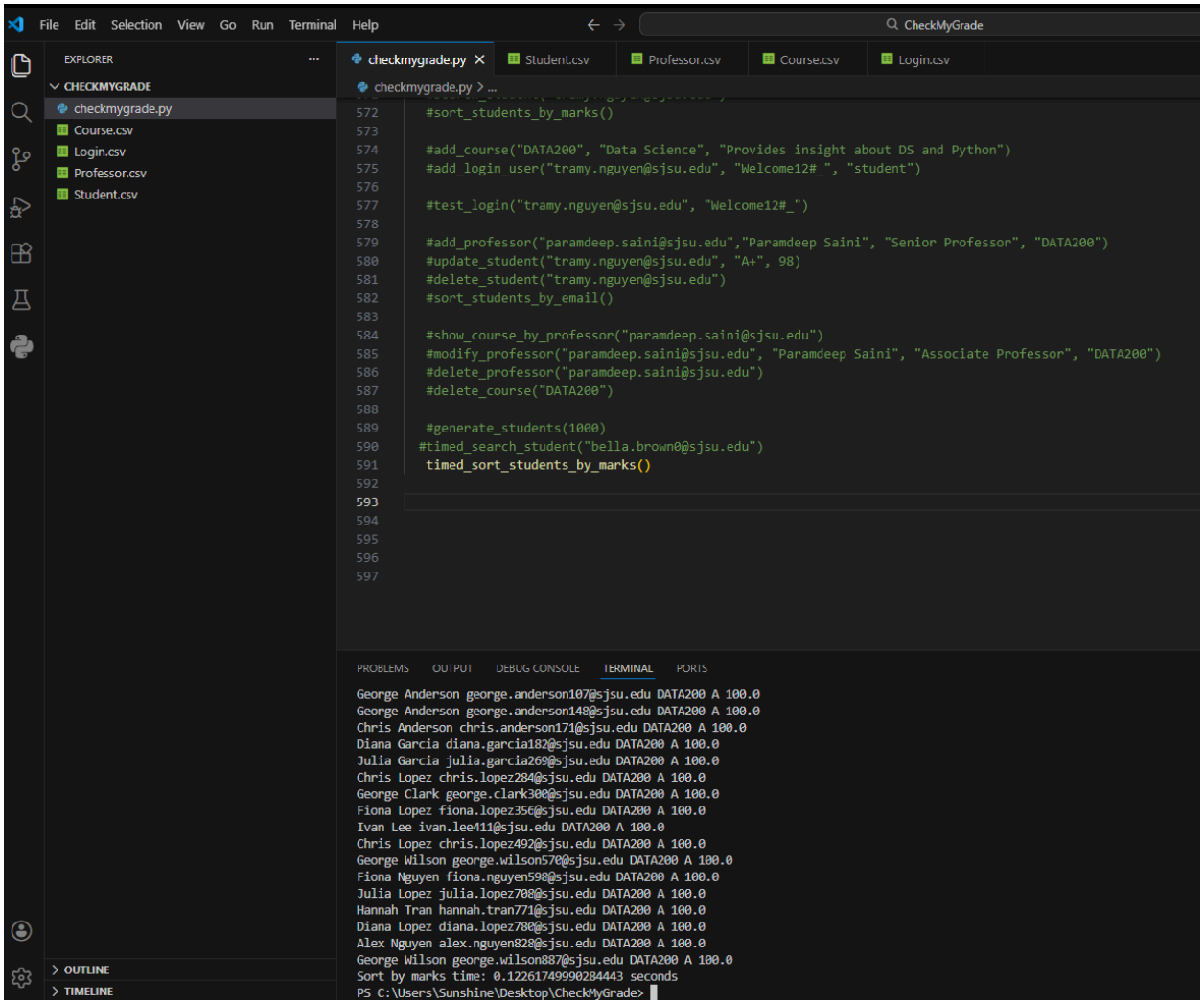
Measuring Search Execution Time: The program calculates and displays the time required to search a student record.



This screenshot shows the search student (bella.brown0@sjsu.edu) and it shows the time search is 0.0030722999…seconds

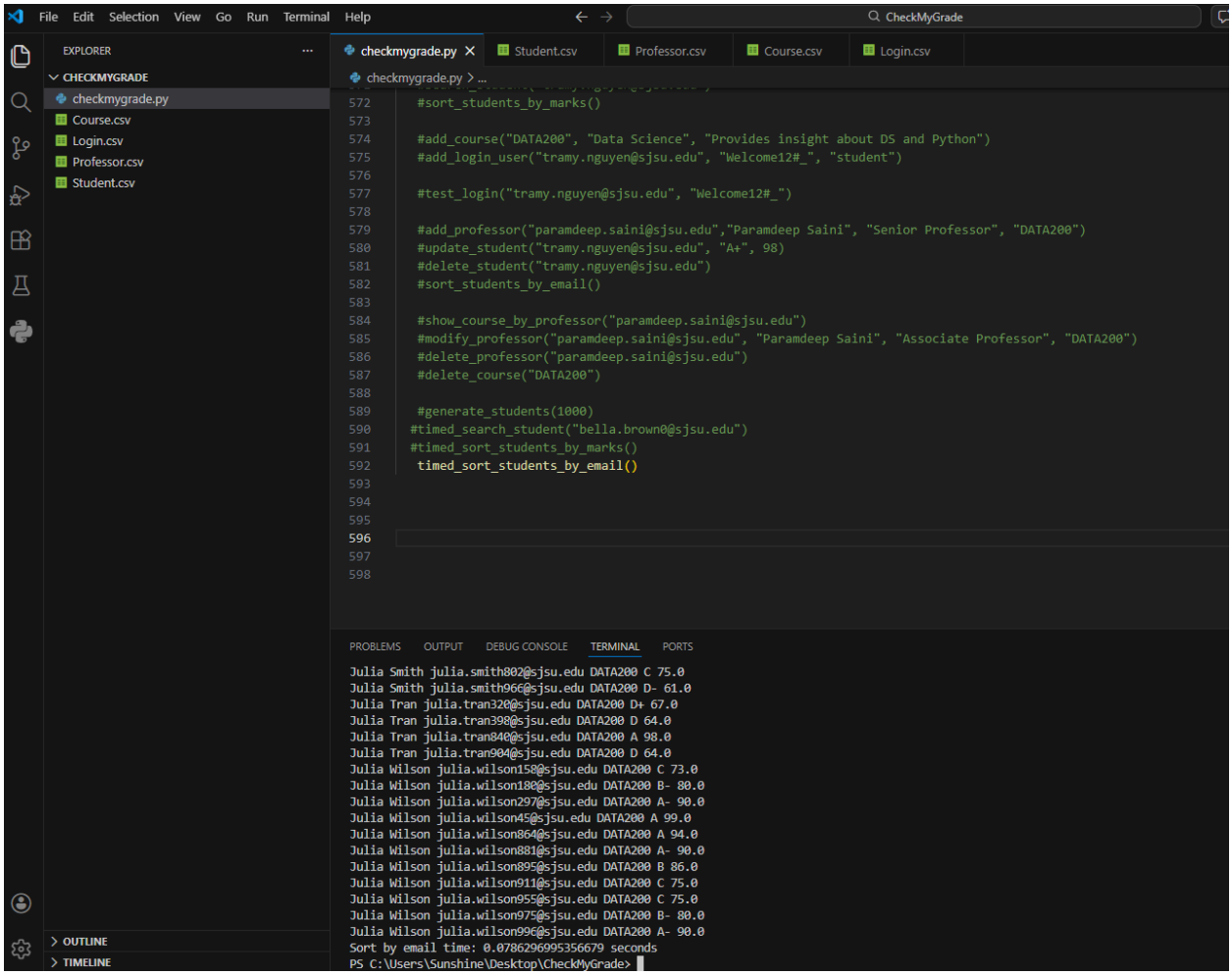
Screenshot Sections

Measuring Sorting Execution Time: The program measures the time required to sort student records by marks or by emails



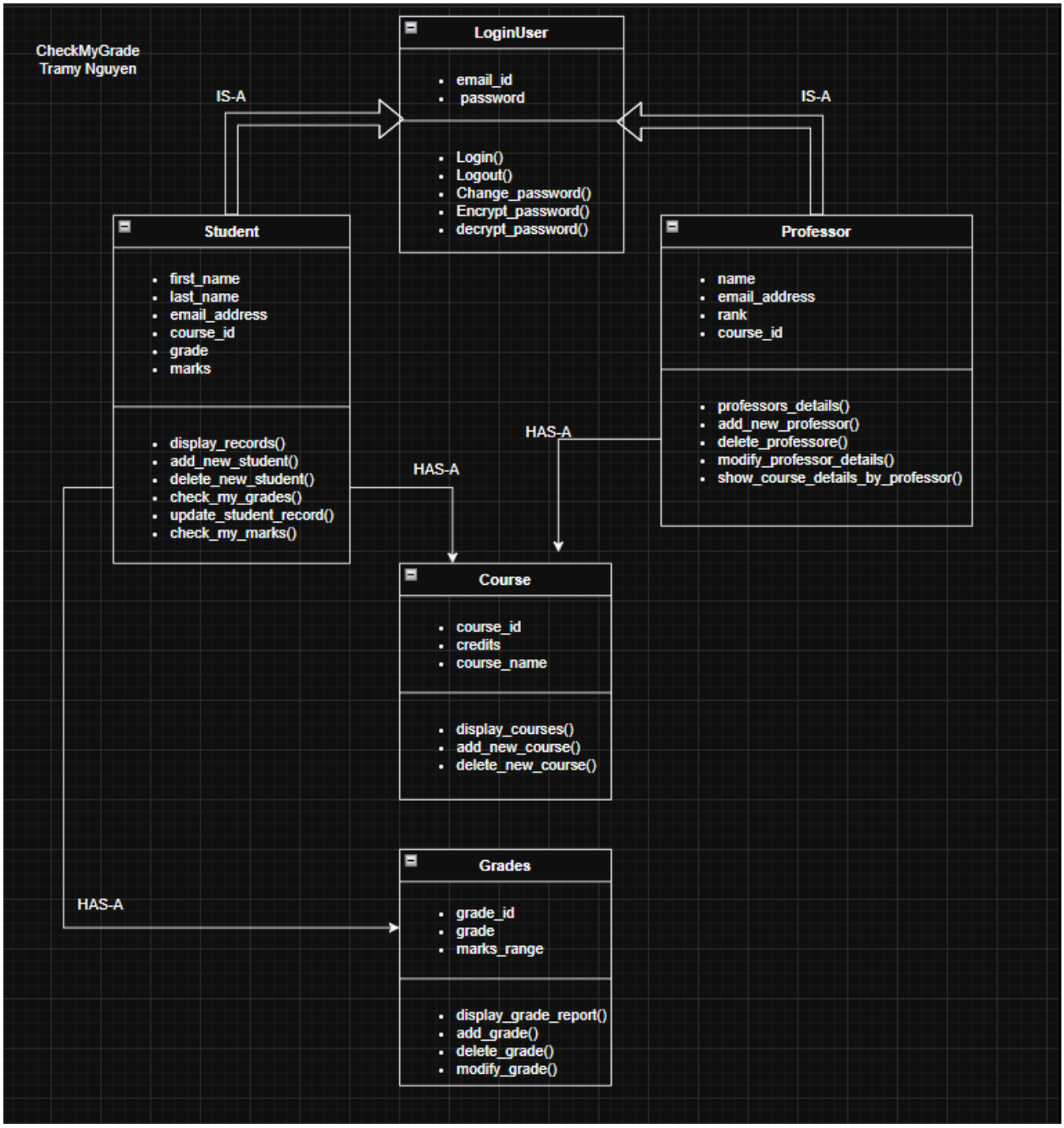
The screenshot shows the time to sort student records by marks, which is around 0.1226174999…seconds

Screenshot Sections



The screenshot shows the time to sort student records by emails, which is around 0.078629699…seconds

OOD Diagram Section



The CheckMyGrade system demonstrates both IS-A and HAS-A relationships. Student and Professor inherit from the LoginUser class (IS-A relationship), as they share authentication functionalities such as login and password management. Additionally, Student has a Course and Grades, and Professor is associated with Course (HAS-A relationships), representing how data is structured and managed within the system.

GitHub URL for the code repository

https://github.com/tramynguyen-22/CheckMyGrade-Application